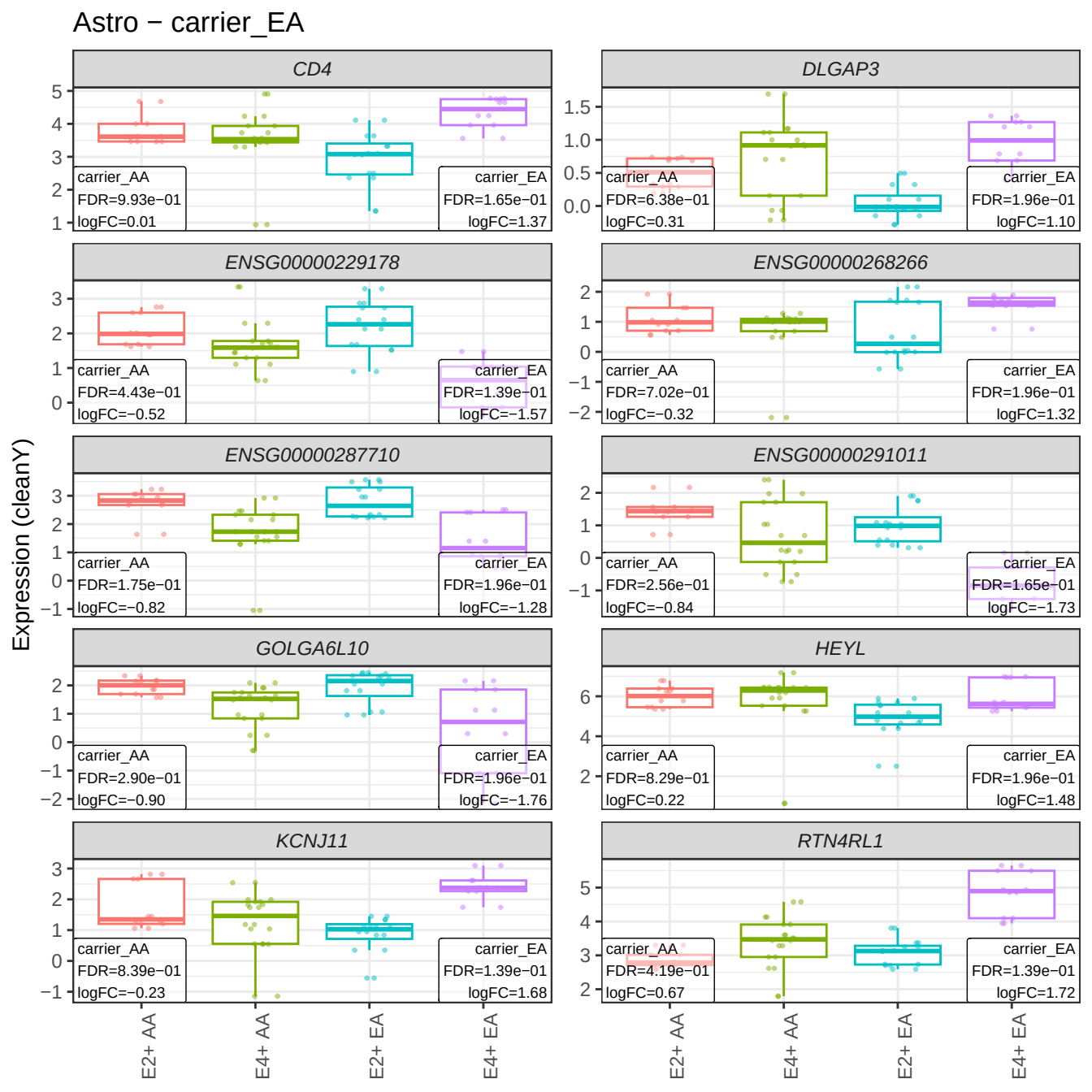
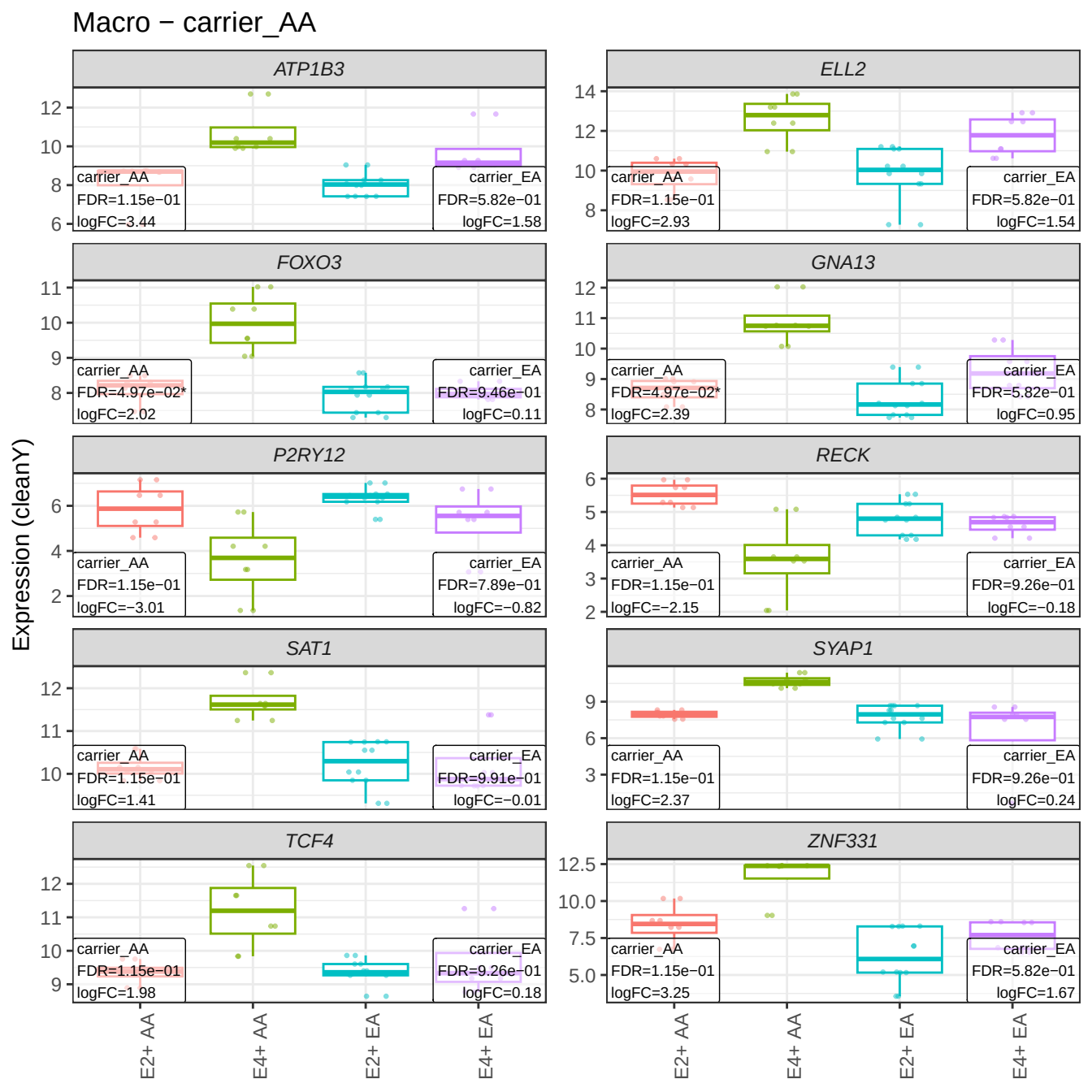
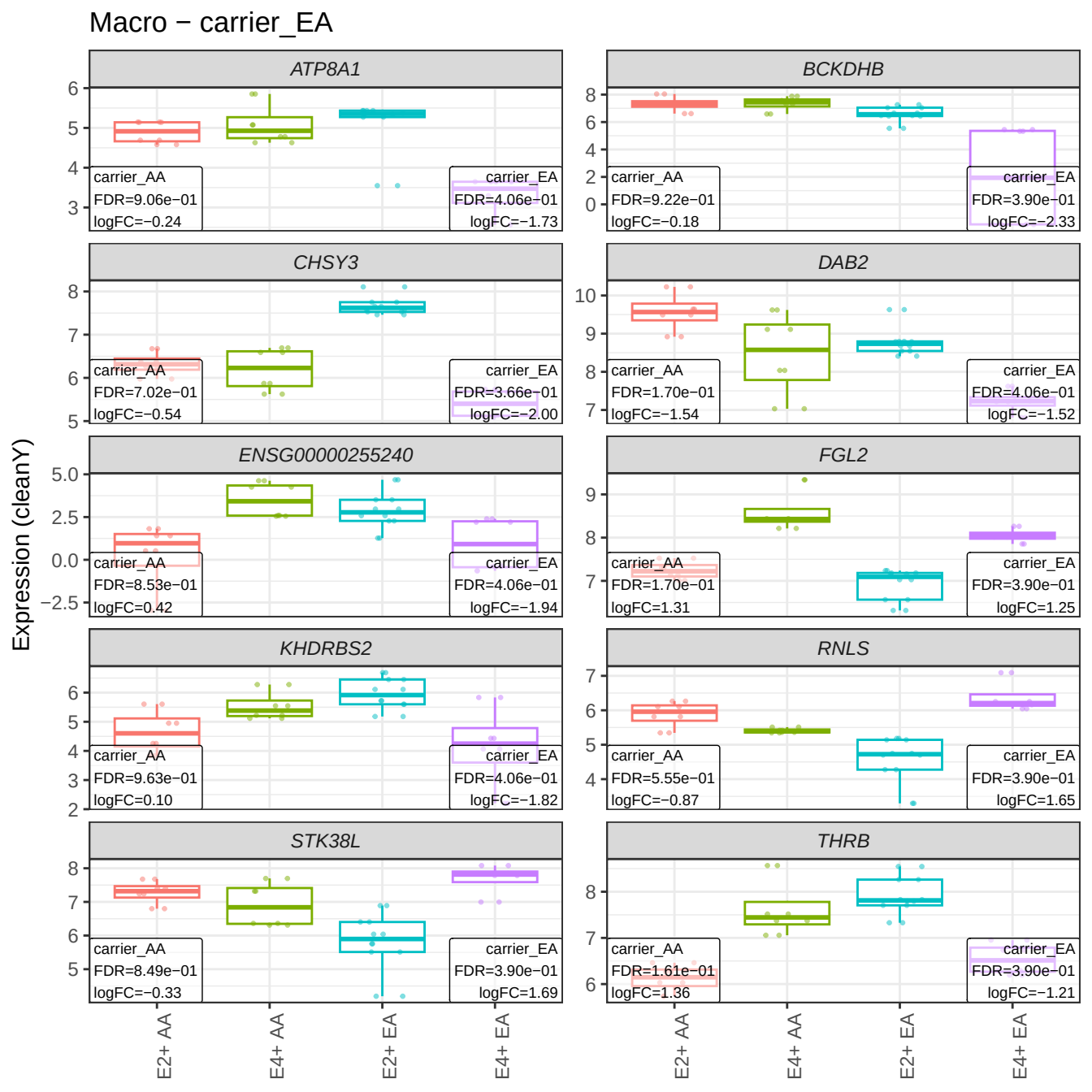
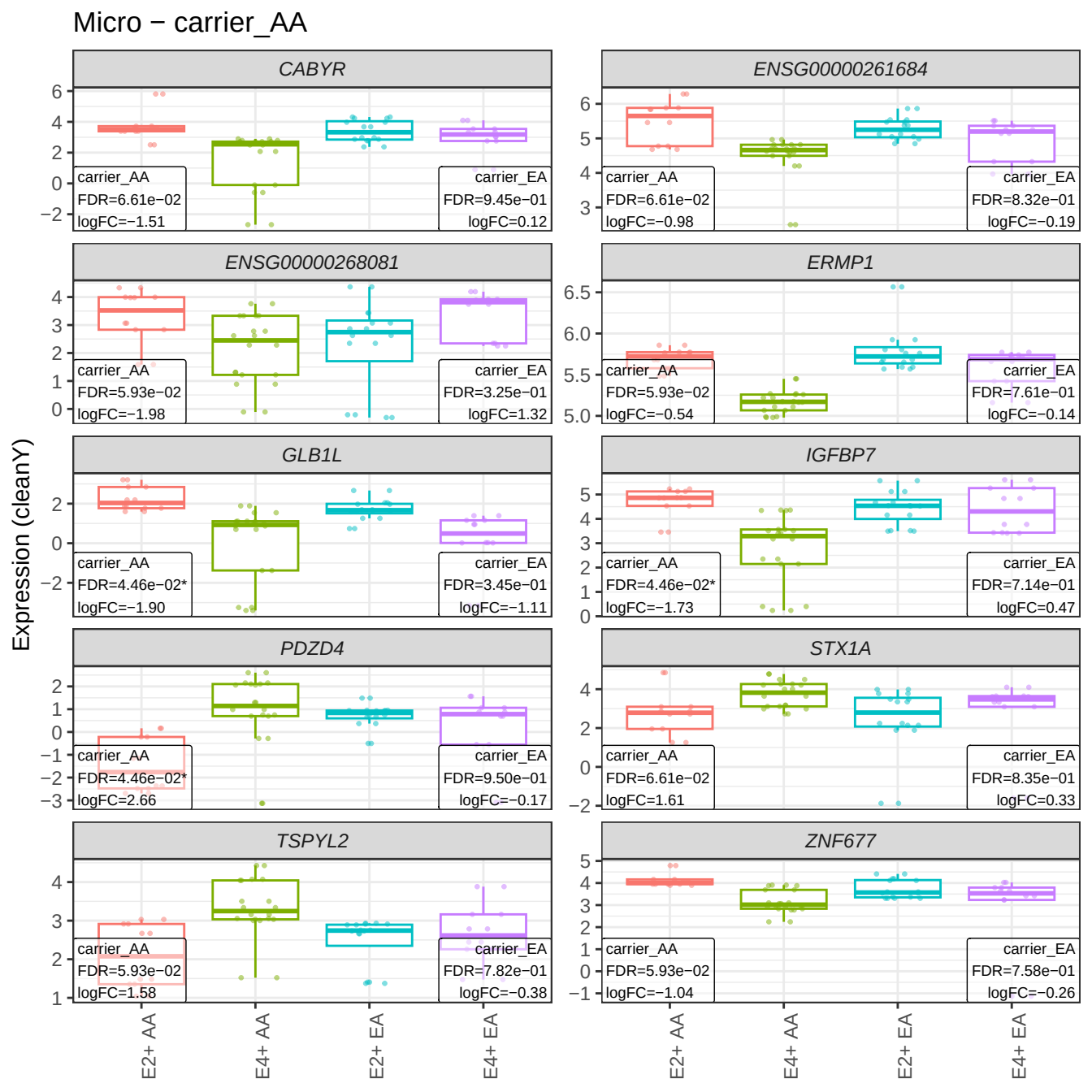
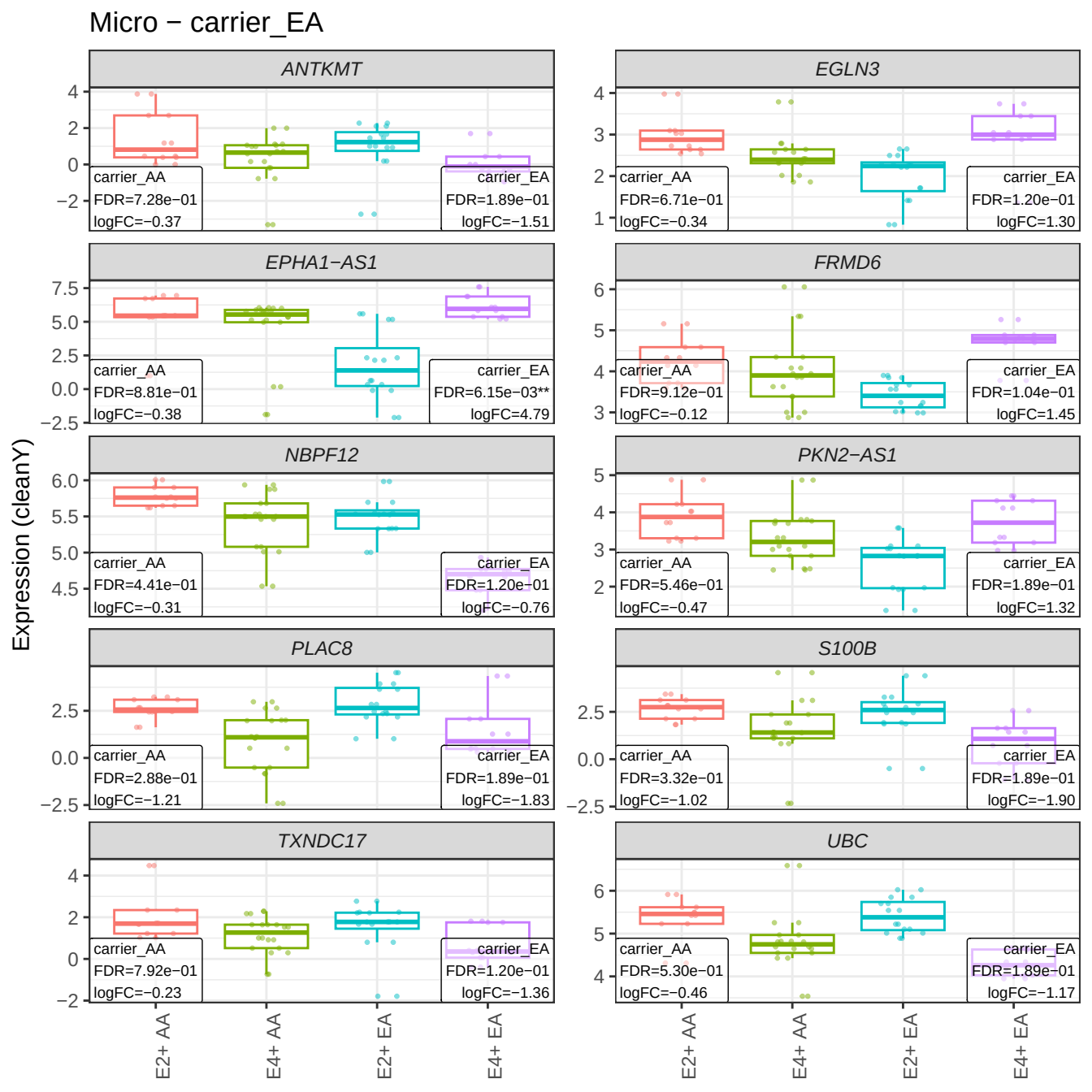
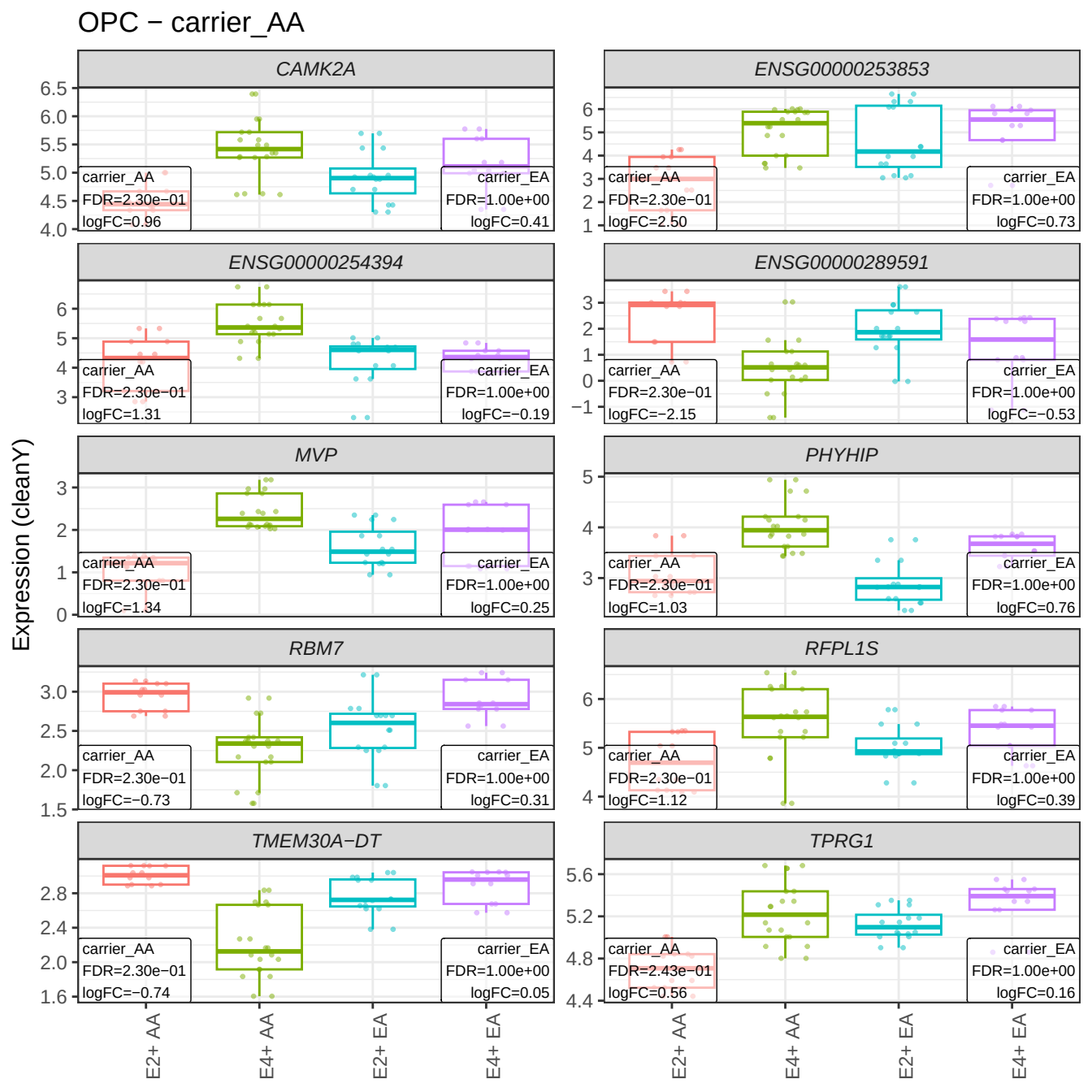

Astro − carrier_EA
CD4
carrier_AA
FDR=9.93e−01
logFC=0.01
carrier_EA
FDR=1.65e−01
logFC=1.37
DLGAP3
carrier_AA
FDR=6.38e−01
logFC=0.31
carrier_EA
FDR=1.96e−01
logFC=1.10
ENSG00000229178
carrier_AA
FDR=4.43e−01
logFC=−0.52
carrier_EA
FDR=1.39e−01
logFC=−1.57
ENSG00000268266
carrier_AA
FDR=7.02e−01
logFC=−0.32
carrier_EA
FDR=1.96e−01
logFC=1.32
ENSG00000287710
carrier_AA
FDR=1.75e−01
logFC=−0.82
carrier_EA
FDR=1.96e−01
logFC=−1.28
ENSG00000291011
carrier_AA
FDR=2.56e−01
logFC=−0.84
carrier_EA
FDR=1.65e−01
logFC=−1.73
GOLGA6L10
carrier_AA
FDR=2.90e−01
logFC=−0.90
carrier_EA
FDR=1.96e−01
logFC=−1.76
HEYL
carrier_AA
FDR=8.29e−01
logFC=0.22
carrier_EA
FDR=1.96e−01
logFC=1.48
KCNJ11
carrier_AA
FDR=8.39e−01
logFC=−0.23
carrier_EA
FDR=1.39e−01
logFC=1.68
RTN4RL1
carrier_AA
FDR=4.19e−01
logFC=0.67
carrier_EA
FDR=1.39e−01
logFC=1.72
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

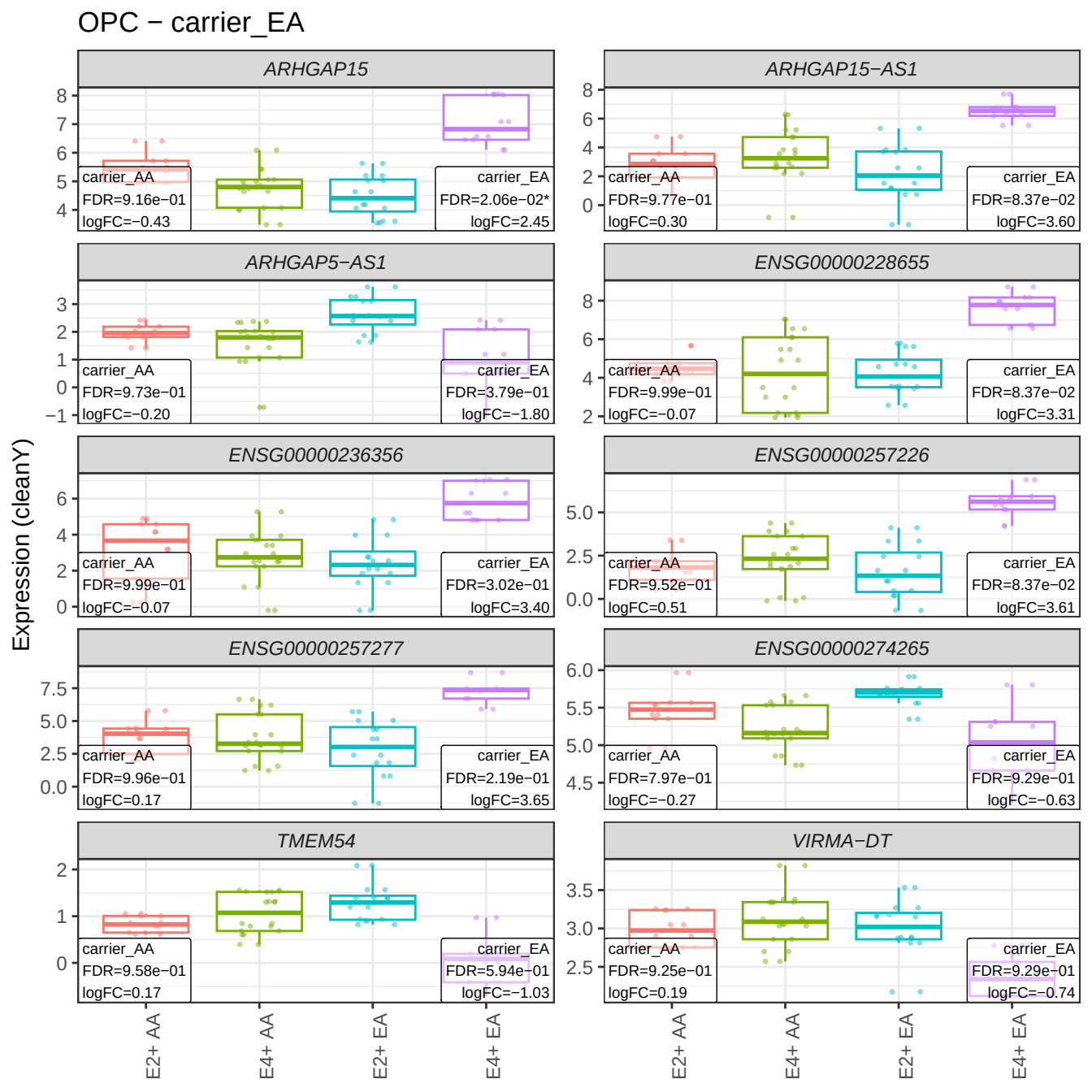

Macro − carrier_AA
ATP1B3
carrier_AA
FDR=1.15e−01
logFC=3.44
carrier_EA
FDR=5.82e−01
logFC=1.58
ELL2
carrier_AA
FDR=1.15e−01
logFC=2.93
carrier_EA
FDR=5.82e−01
logFC=1.54
FOXO3
carrier_AA
FDR=4.97e−02*
logFC=2.02
carrier_EA
FDR=9.46e−01
logFC=0.11
GNA13
carrier_AA
FDR=4.97e−02*
logFC=2.39
carrier_EA
FDR=5.82e−01
logFC=0.95
P2RY12
carrier_AA
FDR=1.15e−01
logFC=−3.01
carrier_EA
FDR=7.89e−01
logFC=−0.82
RECK
carrier_AA
FDR=1.15e−01
logFC=−2.15
carrier_EA
FDR=9.26e−01
logFC=−0.18
SAT1
carrier_AA
FDR=1.15e−01
logFC=1.41
carrier_EA
FDR=9.91e−01
logFC=−0.01
SYAP1
carrier_AA
FDR=1.15e−01
logFC=2.37
carrier_EA
FDR=9.26e−01
logFC=0.24
TCF4
carrier_AA
FDR=1.15e−01
logFC=1.98
carrier_EA
FDR=9.26e−01
logFC=0.18
ZNF331
carrier_AA
FDR=1.15e−01
logFC=3.25
carrier_EA
FDR=5.82e−01
logFC=1.67
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

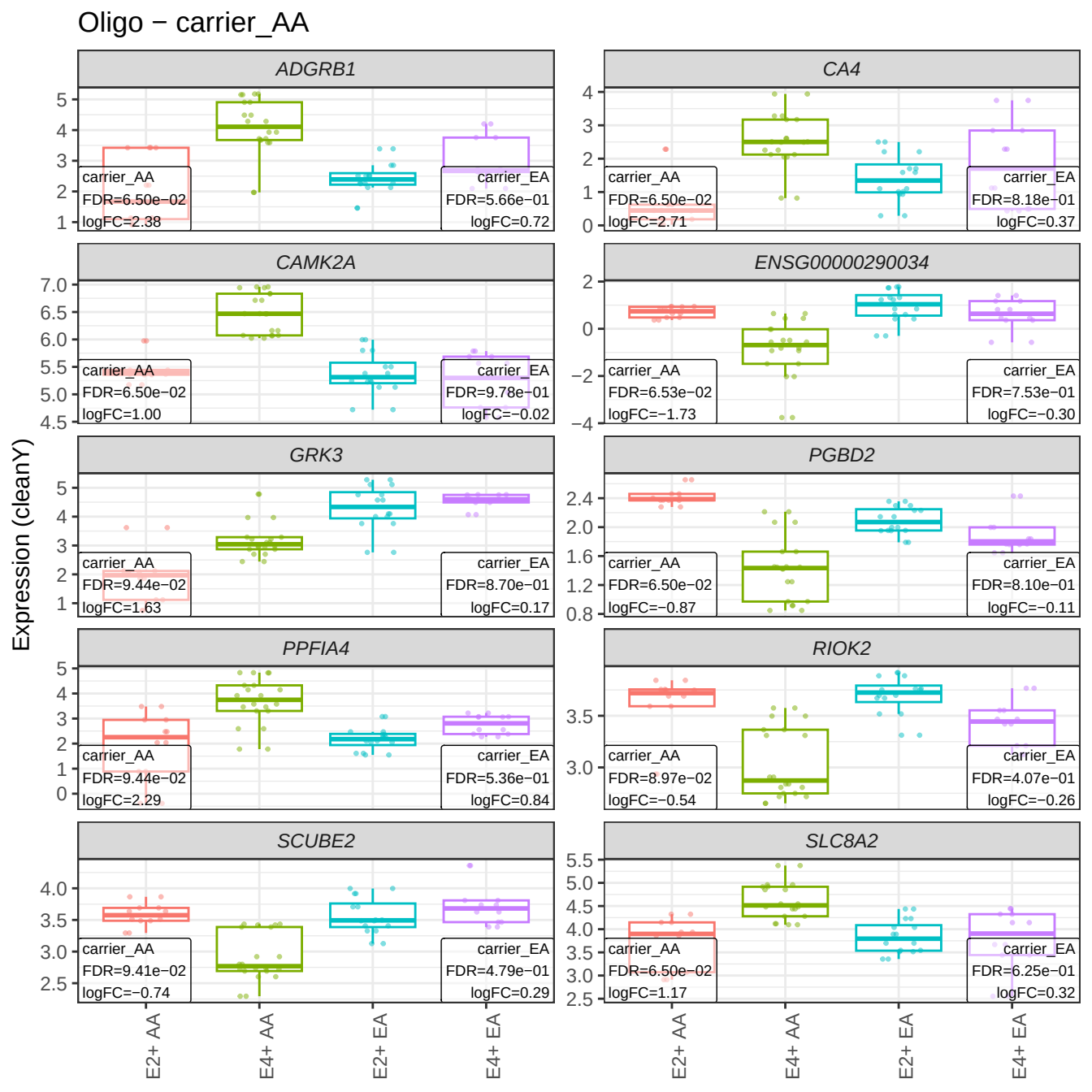

Macro − carrier_EA
ATP8A1
carrier_AA
FDR=9.06e−01
logFC=−0.24
carrier_EA
FDR=4.06e−01
logFC=−1.73
BCKDHB
carrier_AA
FDR=9.22e−01
logFC=−0.18
carrier_EA
FDR=3.90e−01
logFC=−2.33
CHSY3
carrier_AA
FDR=7.02e−01
logFC=−0.54
carrier_EA
FDR=3.66e−01
logFC=−2.00
DAB2
carrier_AA
FDR=1.70e−01
logFC=−1.54
carrier_EA
FDR=4.06e−01
logFC=−1.52
ENSG00000255240
carrier_AA
FDR=8.53e−01
logFC=0.42
carrier_EA
FDR=4.06e−01
logFC=−1.94
FGL2
carrier_AA
FDR=1.70e−01
logFC=1.31
carrier_EA
FDR=3.90e−01
logFC=1.25
KHDRBS2
carrier_AA
FDR=9.63e−01
logFC=0.10
carrier_EA
FDR=4.06e−01
logFC=−1.82
RNLS
carrier_AA
FDR=5.55e−01
logFC=−0.87
carrier_EA
FDR=3.90e−01
logFC=1.65
STK38L
carrier_AA
FDR=8.49e−01
logFC=−0.33
carrier_EA
FDR=3.90e−01
logFC=1.69
THRB
carrier_AA
FDR=1.61e−01
logFC=1.36
carrier_EA
FDR=3.90e−01
logFC=−1.21
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

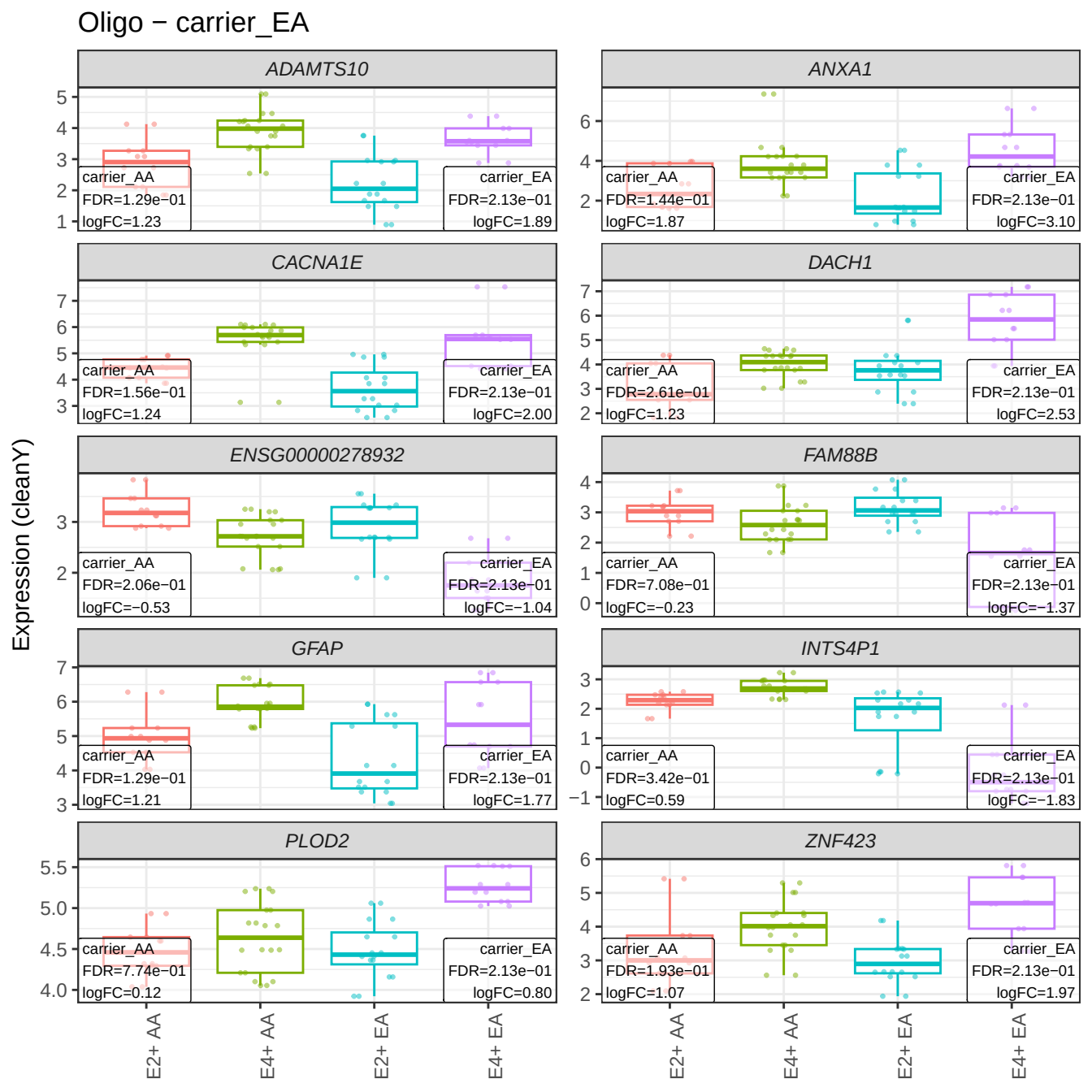

Micro − carrier_AA
CABYR
carrier_AA
FDR=6.61e−02
logFC=−1.51
carrier_EA
FDR=9.45e−01
logFC=0.12
ENSG00000261684
carrier_AA
FDR=6.61e−02
logFC=−0.98
carrier_EA
FDR=8.32e−01
logFC=−0.19
ENSG00000268081
carrier_AA
FDR=5.93e−02
logFC=−1.98
carrier_EA
FDR=3.25e−01
logFC=1.32
ERMP1
carrier_AA
FDR=5.93e−02
logFC=−0.54
carrier_EA
FDR=7.61e−01
logFC=−0.14
GLB1L
carrier_AA
FDR=4.46e−02*
logFC=−1.90
carrier_EA
FDR=3.45e−01
logFC=−1.11
IGFBP7
carrier_AA
FDR=4.46e−02*
logFC=−1.73
carrier_EA
FDR=7.14e−01
logFC=0.47
PDZD4
carrier_AA
FDR=4.46e−02*
logFC=2.66
carrier_EA
FDR=9.50e−01
logFC=−0.17
STX1A
carrier_AA
FDR=6.61e−02
logFC=1.61
carrier_EA
FDR=8.35e−01
logFC=0.33
TSPYL2
carrier_AA
FDR=5.93e−02
logFC=1.58
carrier_EA
FDR=7.82e−01
logFC=−0.38
ZNF677
carrier_AA
FDR=5.93e−02
logFC=−1.04
carrier_EA
FDR=7.58e−01
logFC=−0.26
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

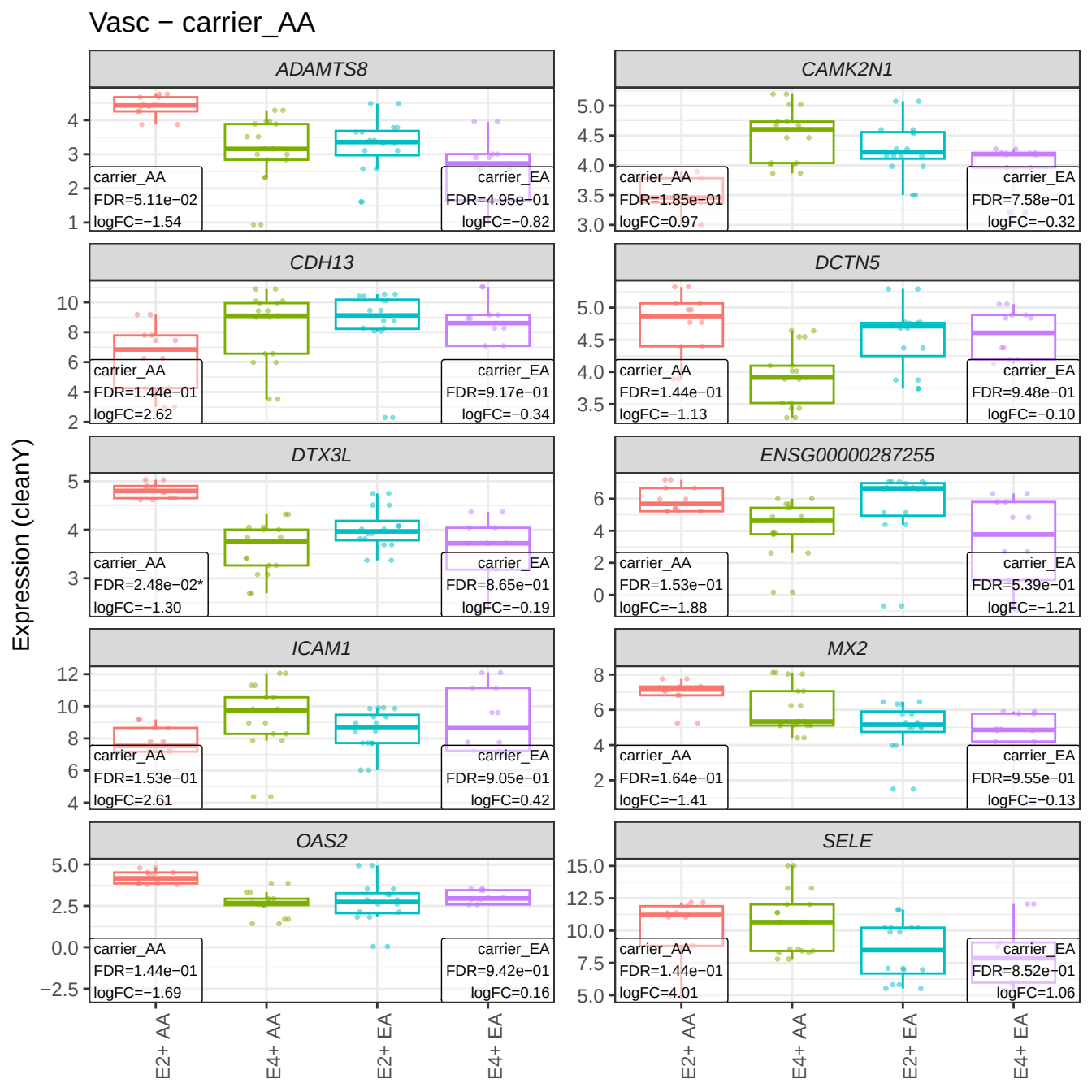

Micro − carrier_EA
Expression (cleanY)
ANTKMT
carrier_AA
FDR=7.28e−01
logFC=−0.37
carrier_EA
FDR=1.89e−01
logFC=−1.51
EGLN3
carrier_AA
FDR=6.71e−01
logFC=−0.34
carrier_EA
FDR=1.20e−01
logFC=1.30
EPHA1−AS1
carrier_AA
FDR=8.81e−01
logFC=−0.38
carrier_EA
FDR=6.15e−03**
logFC=4.79
FRMD6
carrier_AA
FDR=9.12e−01
logFC=−0.12
carrier_EA
FDR=1.04e−01
logFC=1.45
NBPF12
carrier_AA
FDR=4.41e−01
logFC=−0.31
carrier_EA
FDR=1.20e−01
logFC=−0.76
PKN2−AS1
carrier_AA
FDR=5.46e−01
logFC=−0.47
carrier_EA
FDR=1.89e−01
logFC=1.32
PLAC8
carrier_AA
FDR=2.88e−01
logFC=−1.21
carrier_EA
FDR=1.89e−01
logFC=−1.83
S100B
carrier_AA
FDR=3.32e−01
logFC=−1.02
carrier_EA
FDR=1.89e−01
logFC=−1.90
TXNDC17
carrier_AA
FDR=7.92e−01
logFC=−0.23
carrier_EA
FDR=1.20e−01
logFC=−1.36
UBC
carrier_AA
FDR=5.30e−01
logFC=−0.46
carrier_EA
FDR=1.89e−01
logFC=−1.17
E2+ AA
E4+ AA
E2+ EA
E4+ EA

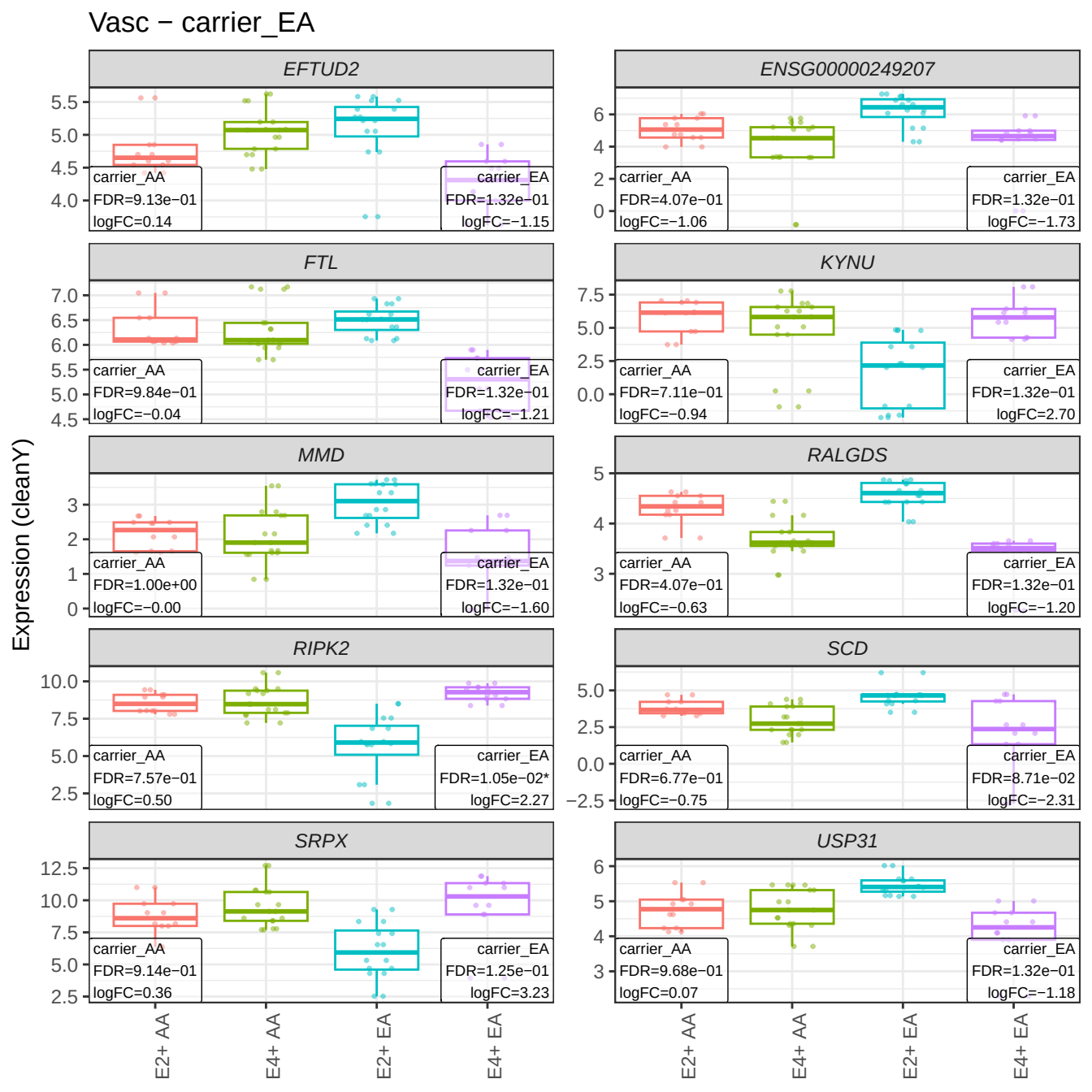

OPC − carrier_AA
CAMK2A
carrier_AA
FDR=2.30e−01
logFC=0.96
carrier_EA
FDR=1.00e+00
logFC=0.41
ENSG00000253853
carrier_AA
FDR=2.30e−01
logFC=2.50
carrier_EA
FDR=1.00e+00
logFC=0.73
ENSG00000254394
carrier_AA
FDR=2.30e−01
logFC=1.31
carrier_EA
FDR=1.00e+00
logFC=−0.19
ENSG00000289591
carrier_AA
FDR=2.30e−01
logFC=−2.15
carrier_EA
FDR=1.00e+00
logFC=−0.53
MVP
carrier_AA
FDR=2.30e−01
logFC=1.34
carrier_EA
FDR=1.00e+00
logFC=0.25
PHYHIP
carrier_AA
FDR=2.30e−01
logFC=1.03
carrier_EA
FDR=1.00e+00
logFC=0.76
RBM7
carrier_AA
FDR=2.30e−01
logFC=−0.73
carrier_EA
FDR=1.00e+00
logFC=0.31
RFPL1S
carrier_AA
FDR=2.30e−01
logFC=1.12
carrier_EA
FDR=1.00e+00
logFC=0.39
TMEM30A−DT
carrier_AA
FDR=2.30e−01
logFC=−0.74
carrier_EA
FDR=1.00e+00
logFC=0.05
TPRG1
carrier_AA
FDR=2.43e−01
logFC=0.56
carrier_EA
FDR=1.00e+00
logFC=0.16
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

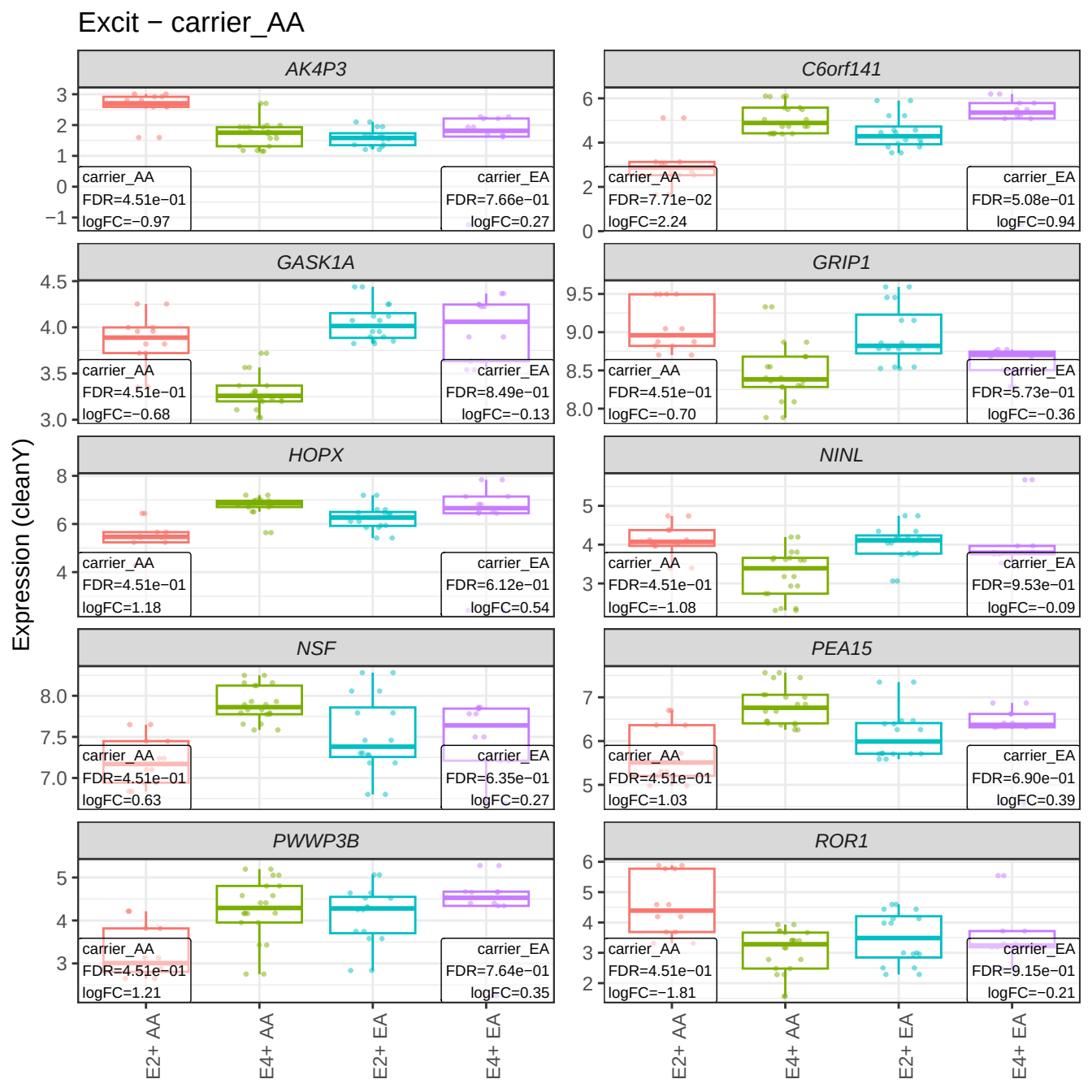

OPC − carrier_EA
ARHGAP15
carrier_AA
FDR=9.16e−01
logFC=−0.43
carrier_EA
FDR=2.06e−02*
logFC=2.45
ARHGAP15−AS1
carrier_AA
FDR=9.77e−01
logFC=0.30
carrier_EA
FDR=8.37e−02
logFC=3.60
ARHGAP5−AS1
carrier_AA
FDR=9.73e−01
logFC=−0.20
carrier_EA
FDR=3.79e−01
logFC=−1.80
ENSG00000228655
carrier_AA
FDR=9.99e−01
logFC=−0.07
carrier_EA
FDR=8.37e−02
logFC=3.31
ENSG00000236356
carrier_AA
FDR=9.99e−01
logFC=−0.07
carrier_EA
FDR=3.02e−01
logFC=3.40
ENSG00000257226
carrier_AA
FDR=9.52e−01
logFC=0.51
carrier_EA
FDR=8.37e−02
logFC=3.61
ENSG00000257277
carrier_AA
FDR=9.96e−01
logFC=0.17
carrier_EA
FDR=2.19e−01
logFC=3.65
ENSG00000274265
carrier_AA
FDR=7.97e−01
logFC=−0.27
carrier_EA
FDR=9.29e−01
logFC=−0.63
TMEM54
carrier_AA
FDR=9.58e−01
logFC=0.17
carrier_EA
FDR=5.94e−01
logFC=−1.03
VIRMA−DT
carrier_AA
FDR=9.25e−01
logFC=0.19
carrier_EA
FDR=9.29e−01
logFC=−0.74
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

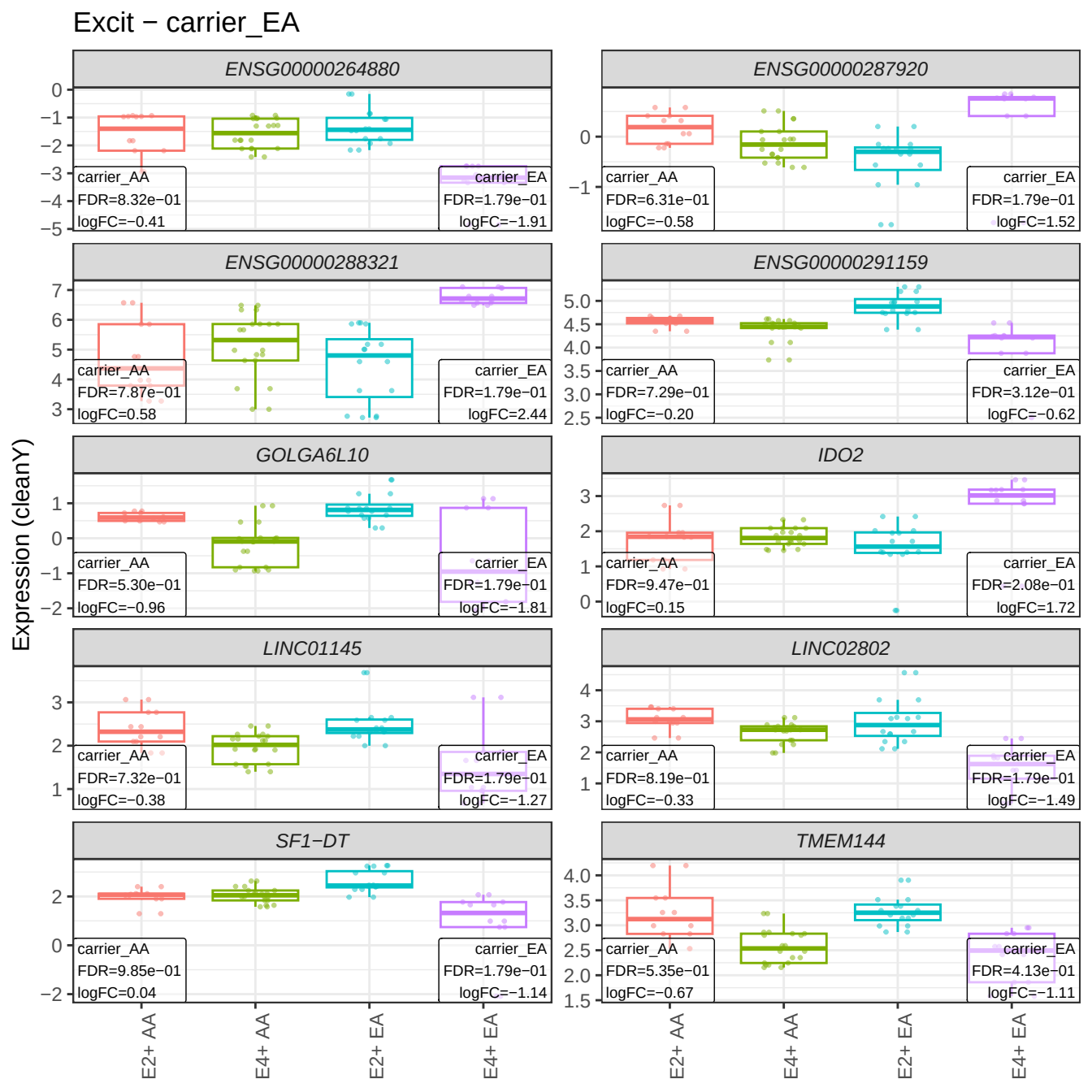

Oligo − carrier_AA
ADGRB1
carrier_AA
FDR=6.50e−02
logFC=2.38
carrier_EA
FDR=5.66e−01
logFC=0.72
CA4
carrier_AA
FDR=6.50e−02
logFC=2.71
carrier_EA
FDR=8.18e−01
logFC=0.37
CAMK2A
carrier_AA
FDR=6.50e−02
logFC=1.00
carrier_EA
FDR=9.78e−01
logFC=−0.02
ENSG00000290034
carrier_AA
FDR=6.53e−02
logFC=−1.73
carrier_EA
FDR=7.53e−01
logFC=−0.30
GRK3
carrier_AA
FDR=9.44e−02
logFC=1.63
carrier_EA
FDR=8.70e−01
logFC=0.17
PGBD2
carrier_AA
FDR=6.50e−02
logFC=−0.87
carrier_EA
FDR=8.10e−01
logFC=−0.11
PPFIA4
carrier_AA
FDR=9.44e−02
logFC=2.29
carrier_EA
FDR=5.36e−01
logFC=0.84
RIOK2
carrier_AA
FDR=8.97e−02
logFC=−0.54
carrier_EA
FDR=4.07e−01
logFC=−0.26
SCUBE2
carrier_AA
FDR=9.41e−02
logFC=−0.74
carrier_EA
FDR=4.79e−01
logFC=0.29
SLC8A2
carrier_AA
FDR=6.50e−02
logFC=1.17
carrier_EA
FDR=6.25e−01
logFC=0.32
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

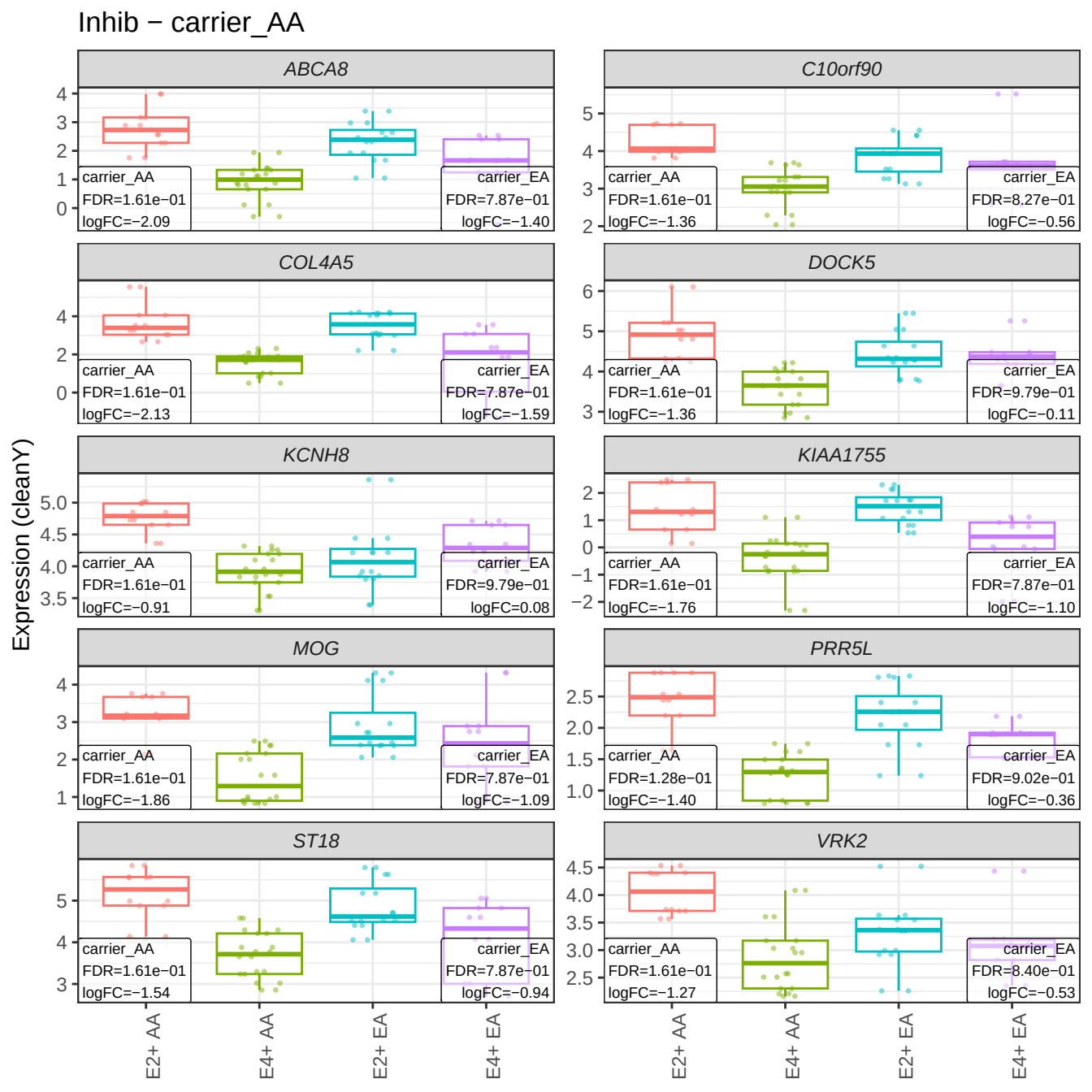

Oligo − carrier_EA
ADAMTS10
carrier_AA
FDR=1.29e−01
logFC=1.23
carrier_EA
FDR=2.13e−01
logFC=1.89
ANXA1
carrier_AA
FDR=1.44e−01
logFC=1.87
carrier_EA
FDR=2.13e−01
logFC=3.10
CACNA1E
carrier_AA
FDR=1.56e−01
logFC=1.24
carrier_EA
FDR=2.13e−01
logFC=2.00
DACH1
carrier_AA
FDR=2.61e−01
logFC=1.23
carrier_EA
FDR=2.13e−01
logFC=2.53
ENSG00000278932
carrier_AA
FDR=2.06e−01
logFC=−0.53
carrier_EA
FDR=2.13e−01
logFC=−1.04
FAM88B
carrier_AA
FDR=7.08e−01
logFC=−0.23
carrier_EA
FDR=2.13e−01
logFC=−1.37
GFAP
carrier_AA
FDR=1.29e−01
logFC=1.21
carrier_EA
FDR=2.13e−01
logFC=1.77
INTS4P1
carrier_AA
FDR=3.42e−01
logFC=0.59
carrier_EA
FDR=2.13e−01
logFC=−1.83
PLOD2
carrier_AA
FDR=7.74e−01
logFC=0.12
carrier_EA
FDR=2.13e−01
logFC=0.80
ZNF423
carrier_AA
FDR=1.93e−01
logFC=1.07
carrier_EA
FDR=2.13e−01
logFC=1.97
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA

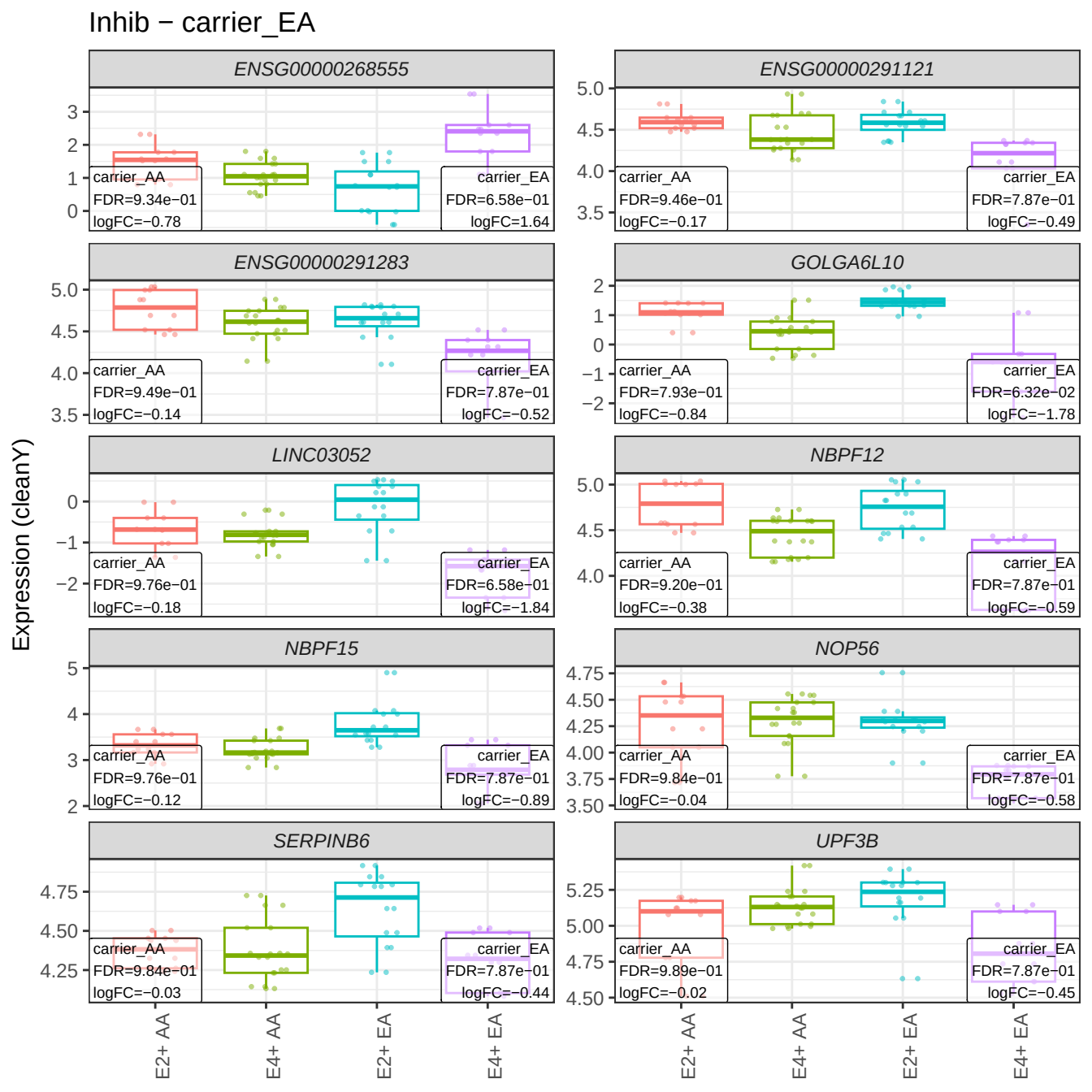

Vasc − carrier_AA
ADAMTS8
carrier_AA
FDR=5.11e−02
logFC=−1.54
carrier_EA
FDR=4.95e−01
logFC=−0.82
CAMK2N1
carrier_AA
FDR=1.85e−01
logFC=0.97
carrier_EA
FDR=7.58e−01
logFC=−0.32
CDH13
carrier_AA
FDR=1.44e−01
logFC=2.62
carrier_EA
FDR=9.17e−01
logFC=−0.34
DCTN5
carrier_AA
FDR=1.44e−01
logFC=−1.13
carrier_EA
FDR=9.48e−01
logFC=−0.10
DTX3L
carrier_AA
FDR=2.48e−02*
logFC=−1.30
carrier_EA
FDR=8.65e−01
logFC=−0.19
ENSG00000287255
carrier_AA
FDR=1.53e−01
logFC=−1.88
carrier_EA
FDR=5.39e−01
logFC=−1.21
ICAM1
carrier_AA
FDR=1.53e−01
logFC=2.61
carrier_EA
FDR=9.05e−01
logFC=0.42
MX2
carrier_AA
FDR=1.64e−01
logFC=−1.41
carrier_EA
FDR=9.55e−01
logFC=−0.13
OAS2
carrier_AA
FDR=1.44e−01
logFC=−1.69
carrier_EA
FDR=9.42e−01
logFC=0.16
SELE
carrier_AA
FDR=1.44e−01
logFC=4.01
carrier_EA
FDR=8.52e−01
logFC=1.06
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA


Vasc − carrier_EA
EFTUD2
carrier_AA
FDR=9.13e−01
logFC=0.14
carrier_EA
FDR=1.32e−01
logFC=−1.15
ENSG00000249207
carrier_AA
FDR=4.07e−01
logFC=−1.06
carrier_EA
FDR=1.32e−01
logFC=−1.73
FTL
carrier_AA
FDR=9.84e−01
logFC=−0.04
carrier_EA
FDR=1.32e−01
logFC=−1.21
KYNU
carrier_AA
FDR=7.11e−01
logFC=−0.94
carrier_EA
FDR=1.32e−01
logFC=2.70
MMD
carrier_AA
FDR=1.00e+00
logFC=−0.00
carrier_EA
FDR=1.32e−01
logFC=−1.60
RALGDS
carrier_AA
FDR=4.07e−01
logFC=−0.63
carrier_EA
FDR=1.32e−01
logFC=−1.20
RIPK2
carrier_AA
FDR=7.57e−01
logFC=0.50
carrier_EA
FDR=1.05e−02*
logFC=2.27
SCD
carrier_AA
FDR=6.77e−01
logFC=−0.75
carrier_EA
FDR=8.71e−02
logFC=−2.31
SRPX
carrier_AA
FDR=9.14e−01
logFC=0.36
carrier_EA
FDR=1.25e−01
logFC=3.23
USP31
carrier_AA
FDR=9.68e−01
logFC=0.07
carrier_EA
FDR=1.32e−01
logFC=−1.18
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA


Excit − carrier_AA
AK4P3
carrier_AA
FDR=4.51e−01
logFC=−0.97
carrier_EA
FDR=7.66e−01
logFC=0.27
C6orf141
carrier_AA
FDR=7.71e−02
logFC=2.24
carrier_EA
FDR=5.08e−01
logFC=0.94
GASK1A
carrier_AA
FDR=4.51e−01
logFC=−0.68
carrier_EA
FDR=8.49e−01
logFC=−0.13
GRIP1
carrier_AA
FDR=4.51e−01
logFC=−0.70
carrier_EA
FDR=5.73e−01
logFC=−0.36
HOPX
carrier_AA
FDR=4.51e−01
logFC=1.18
carrier_EA
FDR=6.12e−01
logFC=0.54
NINL
carrier_AA
FDR=4.51e−01
logFC=−1.08
carrier_EA
FDR=9.53e−01
logFC=−0.09
NSF
carrier_AA
FDR=4.51e−01
logFC=0.63
carrier_EA
FDR=6.35e−01
logFC=0.27
PEA15
carrier_AA
FDR=4.51e−01
logFC=1.03
carrier_EA
FDR=6.90e−01
logFC=0.39
PWWP3B
carrier_AA
FDR=4.51e−01
logFC=1.21
carrier_EA
FDR=7.64e−01
logFC=0.35
ROR1
carrier_AA
FDR=4.51e−01
logFC=−1.81
carrier_EA
FDR=9.15e−01
logFC=−0.21
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA


Excit − carrier_EA
ENSG00000264880
carrier_AA
FDR=8.32e−01
logFC=−0.41
carrier_EA
FDR=1.79e−01
logFC=−1.91
ENSG00000287920
carrier_AA
FDR=6.31e−01
logFC=−0.58
carrier_EA
FDR=1.79e−01
logFC=1.52
ENSG00000288321
carrier_AA
FDR=7.87e−01
logFC=0.58
carrier_EA
FDR=1.79e−01
logFC=2.44
ENSG00000291159
carrier_AA
FDR=7.29e−01
logFC=−0.20
carrier_EA
FDR=3.12e−01
logFC=−0.62
GOLGA6L10
carrier_AA
FDR=5.30e−01
logFC=−0.96
carrier_EA
FDR=1.79e−01
logFC=−1.81
IDO2
carrier_AA
FDR=9.47e−01
logFC=0.15
carrier_EA
FDR=2.08e−01
logFC=1.72
LINC01145
carrier_AA
FDR=7.32e−01
logFC=−0.38
carrier_EA
FDR=1.79e−01
logFC=−1.27
LINC02802
carrier_AA
FDR=8.19e−01
logFC=−0.33
carrier_EA
FDR=1.79e−01
logFC=−1.49
SF1−DT
carrier_AA
FDR=9.85e−01
logFC=0.04
carrier_EA
FDR=1.79e−01
logFC=−1.14
TMEM144
carrier_AA
FDR=5.35e−01
logFC=−0.67
carrier_EA
FDR=4.13e−01
logFC=−1.11
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA


Inhib − carrier_AA
ABCA8
carrier_AA
FDR=1.61e−01
logFC=−2.09
carrier_EA
FDR=7.87e−01
logFC=−1.40
C10orf90
carrier_AA
FDR=1.61e−01
logFC=−1.36
carrier_EA
FDR=8.27e−01
logFC=−0.56
COL4A5
carrier_AA
FDR=1.61e−01
logFC=−2.13
carrier_EA
FDR=7.87e−01
logFC=−1.59
DOCK5
carrier_AA
FDR=1.61e−01
logFC=−1.36
carrier_EA
FDR=9.79e−01
logFC=−0.11
KCNH8
carrier_AA
FDR=1.61e−01
logFC=−0.91
carrier_EA
FDR=9.79e−01
logFC=0.08
KIAA1755
carrier_AA
FDR=1.61e−01
logFC=−1.76
carrier_EA
FDR=7.87e−01
logFC=−1.10
MOG
carrier_AA
FDR=1.61e−01
logFC=−1.86
carrier_EA
FDR=7.87e−01
logFC=−1.09
PRR5L
carrier_AA
FDR=1.28e−01
logFC=−1.40
carrier_EA
FDR=9.02e−01
logFC=−0.36
ST18
carrier_AA
FDR=1.61e−01
logFC=−1.54
carrier_EA
FDR=7.87e−01
logFC=−0.94
VRK2
carrier_AA
FDR=1.61e−01
logFC=−1.27
carrier_EA
FDR=8.40e−01
logFC=−0.53
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA


Inhib − carrier_EA
ENSG00000268555
carrier_AA
FDR=9.34e−01
logFC=−0.78
carrier_EA
FDR=6.58e−01
logFC=1.64
ENSG00000291121
carrier_AA
FDR=9.46e−01
logFC=−0.17
carrier_EA
FDR=7.87e−01
logFC=−0.49
ENSG00000291283
carrier_AA
FDR=9.49e−01
logFC=−0.14
carrier_EA
FDR=7.87e−01
logFC=−0.52
GOLGA6L10
carrier_AA
FDR=7.93e−01
logFC=−0.84
carrier_EA
FDR=6.32e−02
logFC=−1.78
LINC03052
carrier_AA
FDR=9.76e−01
logFC=−0.18
carrier_EA
FDR=6.58e−01
logFC=−1.84
NBPF12
carrier_AA
FDR=9.20e−01
logFC=−0.38
carrier_EA
FDR=7.87e−01
logFC=−0.59
NBPF15
carrier_AA
FDR=9.76e−01
logFC=−0.12
carrier_EA
FDR=7.87e−01
logFC=−0.89
NOP56
carrier_AA
FDR=9.84e−01
logFC=−0.04
carrier_EA
FDR=7.87e−01
logFC=−0.58
SERPINB6
carrier_AA
FDR=9.84e−01
logFC=−0.03
carrier_EA
FDR=7.87e−01
logFC=−0.44
UPF3B
carrier_AA
FDR=9.89e−01
logFC=−0.02
carrier_EA
FDR=7.87e−01
logFC=−0.45
Expression (cleanY)
E2+ AA
E4+ AA
E2+ EA
E4+ EA